



## CASE STUDY

# AI in Beauty Products Manufacturing and Supply Chain

Innovacio unified ERP, MES, WMS, QMS, supplier, and channel data into one AI command center for a beauty manufacturer, using 27-product demand modeling, predictive maintenance, and vision-based quality control to reduce overproduction, protect traceability, and improve launch and packaging reliability.

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## Problem

The client's beauty manufacturing and supply chain operations were spread across ERP, MES, WMS, QMS, supplier records, and channel data, which made it difficult to manage demand, sourcing, production, quality, and compliance as one connected flow. Planning teams were dealing with too many SKUs, frequent launches, and trend-driven demand swings, while procurement and warehouse teams lacked clear visibility into raw-material needs, stock risk, and ingredient traceability.

Formulation work was also slow because ingredient performance, supplier quality, stability results, and past batch outcomes were scattered across different systems and documents. On the production side, mixing, filling, capping, and packaging lines had limited predictive visibility, so downtime and batch interruptions were often handled reactively. Packaging defects, label issues, fill-level drift, and traceability gaps created additional risk for returns, brand damage, and compliance issues.

## How we solved it

Innovacio Technologies designed an AI-powered operating layer that unified planning, formulation, sourcing, production, quality, and traceability data into one live command center. This gave the client a single operational view instead of fragmented reports and manual coordination across departments.

We implemented demand forecasting to improve SKU-level planning using sales history, promotions, review signals, and lifecycle patterns, helping the client reduce overproduction and stockouts. We also built formulation intelligence to support ingredient selection, performance prediction, hazard screening, and smarter raw-material decisions based on lab data, supplier quality, and historical batch performance.

To improve reliability, we added predictive-maintenance capabilities for critical equipment such as mixers, fillers, cappers, and packaging assets so issues could be flagged earlier and line stoppages reduced. We also deployed visual inspection and traceability workflows to catch packaging defects, label errors, and dimensional issues while linking raw materials to batch records, certificates, and finished goods for stronger compliance and recall readiness.



The result was a more connected and scalable beauty operations model where planning, quality, production, and sourcing teams could work from the same data foundation, make faster decisions, and protect both product quality and brand trust.

## EXECUTIVE SUMMARY

Beauty manufacturing looks simple from the outside, but the operating reality is complex: product lines change quickly, raw materials come from many sources, shelf-life is limited, and consumer demand shifts with trends, seasonality, influencers, and promotions. A plant-based beauty manufacturer has to balance clean-label expectations with reliable production, traceable sourcing, and responsive fulfillment.

AI creates value in this environment by turning disconnected information into one operating picture. It can forecast demand at SKU level, flag supply risk early, optimize formulations and ingredient choices, detect packaging defects before shipment, and help procurement and quality teams work from the same live truth.

This case study translates those research findings into a practical Innovacio-style story: what was broken, what data existed, what AI layer we built, and what operational improvement followed.

## WHY BEAUTY MANUFACTURING IS DIFFERENT

- High SKU variety: shades, scents, sizes, bundles, seasonal launches, and channel-specific packs create short product life cycles and frequent planning changes.
- Ingredient sensitivity: formulas depend on critical materials such as surfactants, polymers, fragrances, preservatives, hydrogels, oils, and botanical inputs, which need careful compatibility and quality control.
- Traceability pressure: plant-based and clean beauty claims depend on source transparency, batch traceability, and reliable quality verification.
- Packaging and appearance matter: fill level, cap torque, seal integrity, label placement, and cosmetic appearance all affect sell-through and returns.
- Demand is trend-driven: social media, reviews, promotions, and macro shocks can distort forecasts much faster than in slower-moving categories.

| Beauty AI map       | What the team needs                          | What AI adds                              | Operational effect              |
|---------------------|----------------------------------------------|-------------------------------------------|---------------------------------|
| Demand planning     | Channel sales, reviews, promotions, launches | SKU forecasting and bias correction       | Fewer stock-outs and overstocks |
| Formulation         | Ingredient, stability, and performance data  | Formula optimization and hazard screening | Faster product development      |
| Supply traceability | Supplier, batch, and certification data      | Risk scoring and traceability links       | Cleaner compliance and sourcing |

|         |                                       |                                         |                               |
|---------|---------------------------------------|-----------------------------------------|-------------------------------|
| Quality | Images, defect labels, fill/pack data | Visual inspection and anomaly detection | Lower escape and rework rates |
|---------|---------------------------------------|-----------------------------------------|-------------------------------|

## DEMAND FORECASTING AND PRODUCTION PLANNING

**Business problem:** The manufacturer was carrying the classic beauty-industry burden: too many SKUs, too many launch variants, and too much dependency on intuition-based planning. Some products were fast-ramping and trend-sensitive, while others were slow movers with long tails. That made it hard to know what to produce, when to buy ingredients, and how much finished inventory to hold.

**Data problem:** The useful signals already existed, but they were scattered. Sales history lived in ERP and channel systems, online reviews sat in e-commerce platforms, promotion data was in marketing calendars, and demand signals from wholesale, DTC, and marketplace channels were not reconciled into one planning view.

**AI solution:** Innovacio built a demand-planning layer that combined SKU-level sales history, promotion intensity, review sentiment, product lifecycle stage, and channel mix into one forecasting engine. The design followed the literature's guidance that beauty forecasts improve when managers use longer-horizon and product-characteristic-sensitive models rather than intuition alone. In the 2020 beauty-industry study, four fuzzy inference system models were tested on 27 nail-polish products from a multinational beauty company, and the authors found that low-kurtosis, negatively skewed time series and longer-horizon forecasts were best served by the fuzzy models rather than informal qualitative planning.

**Measured evidence from the literature:** Another beauty-care study shows that pandemic shocks and macroeconomic behavior changes can weaken AI forecasting accuracy, which means the forecasting layer must be monitored and refreshed rather than treated as static. Taken together, the research supports a hybrid planning approach: AI forecasts, planners validate, and exceptions are handled by humans.

| Evidence                | What the study found                                                            | Why it matters for beauty operations                              |
|-------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------|
| 27-product beauty case  | Fuzzy inference models tested on a nail-polish line                             | Matches the short-life-cycle, trend-driven reality of beauty SKUs |
| Forecast selection rule | Best for low-kurtosis, negatively skewed, longer-horizon demand                 | Useful for launch items and ramping products                      |
| Pandemic shock study    | AI forecast accuracy was affected by macro shocks and consumer behavior changes | Forecasts must be monitored, not assumed stable                   |





- Build a knowledge base of past formula trials, including failed batches, so the model learns from both success and failure.
- Tie ingredient selection to procurement and supplier quality data, not only to lab performance.
- Use explainable ranking outputs so chemists and QA teams can see why an ingredient combination is recommended.
- Re-score key materials when supplier origin, certification, or purity changes.

## PREDICTIVE MAINTENANCE FOR MIXING, FILLING, AND PACKAGING LINES

**Business problem:** In beauty manufacturing, downtime is expensive because mixing vessels, homogenizers, pumps, fillers, cappers, labelers, and cartoners all sit in a tightly sequenced line. A single failure can stop production, delay launches, and create waste in semi-finished product.

**Data problem:** The signals are usually there but not connected: vibration, temperature, torque, pressure, cycle counts, alarms, work orders, and downtime notes. Without integration, maintenance is reactive and teams only see the problem after a line stop.

**AI solution:** Innovacio built a predictive-maintenance layer that learned from equipment sensors, maintenance records, and stoppage history. The model scored asset health, flagged anomaly patterns early, and recommended intervention windows during planned downtime. That approach matches the broader predictive-maintenance literature showing that AI and IoT are key enablers for more efficient manufacturing maintenance, especially when models are linked to live process data.

**Impact logic:** The operational gain is fewer emergency interventions, lower scrap from interrupted batches, better spare-parts planning, and stronger schedule stability. In a beauty plant, that also protects filling consistency, packaging quality, and order fulfillment.

- Start with the most critical assets: mixers, homogenizers, fillers, cappers, and packaging cells.
- Reuse existing PLC and maintenance data before adding expensive hardware.
- Keep the model explainable enough for maintenance and operations leaders to trust the alert.
- Revalidate the model when a line is changed, a new SKU is introduced, or a major service event occurs.

## PACKAGING INSPECTION, TRACEABILITY, AND COMPLIANCE

**Business problem:** Appearance defects are not cosmetic in the business sense; they are operational defects. Mislabels, fill-level drift, damaged seals, contamination risk, and packaging scratches can lead to returns, complaints, and brand damage. At the same time, plant-based and clean-beauty positioning increases the need for reliable batch traceability and source verification.

**Data problem:** Packaging defects are visual and fast-moving, while traceability evidence is dispersed across suppliers, quality tests, warehouse receipts, and batch records. Manual review is too slow and too inconsistent for modern output speeds.



AI solution: Innovacio built a combined visual-QC and traceability workflow. On the quality side, the system used computer vision to inspect labels, seals, fill level, and surface defects; on the compliance side, it linked raw materials to batch records, certificates, and finished goods. Recent inspection research shows the value of exactly this kind of system: a multi-parameter inspection platform for transparent packaging containers integrates stress measurement, dimensional measurement, and defect detection in one architecture, reuses shared optics to avoid extra hardware, and sends results to MES for traceability and quality management. The same study reports dimensional errors within +/-0.2 mm and high repeatability under production-like conditions.

The supply-chain angle matters too. A cosmetics-supply-chain study on Thailand found that the chain remains opaque and suggested a role for independent quality assurance verifiers and traceability structures to support high-trust cosmetic markets. That aligns with the needs of a plant-based beauty brand, where source transparency is part of the value proposition.

| Quality / traceability       | AI contribution                                      | Business outcome                     |
|------------------------------|------------------------------------------------------|--------------------------------------|
| Packaging inspection         | Detects defects, label issues, and dimensional drift | Fewer returns and escapes            |
| Stress and dimension control | Checks container integrity and geometry              | Better consistency in shipping units |
| Traceability                 | Links batches, suppliers, and certificates           | Faster recalls and cleaner audits    |
| Quality assurance verifiers  | Adds structured verification to supplier flows       | Higher trust in sourcing claims      |

Suggested implementation details:

- Inspect packaging at line speed and keep the defect taxonomy simple enough for operators to use.
- Tie every finished batch to the raw-material batches that were actually consumed.
- Build exception workflows for suspect lots, certificate gaps, and supplier changes.
- Keep human review on the final disposition for any critical quality or compliance event.

## QUALITY, SUSTAINABILITY, AND COMPLIANCE LAYER

AI is most credible in beauty manufacturing when it reduces manual drudgery and strengthens confidence in the product rather than pretending to replace responsible experts. The best use cases are formula screening, demand planning, packaging QC, source traceability, and exception triage. AI also supports sustainability by reducing waste, improving planning, and helping teams avoid overproduction of short-life SKUs.

For a plant-based beauty company, this matters because customers expect clean sourcing, responsible manufacturing, and honest claims. AI helps the organization prove those claims with data.

- Formula review by exception



- Supplier risk scoring
- Batch and lot traceability
- Packaging defect triage
- Promotion and launch planning

## RECOMMENDED ARCHITECTURE FOR INNOVACIO

A strong Innovacio deployment should show a practical operating model, not just a model score. The right structure is a governed data layer, an AI layer, a workflow layer, and a governance layer.

| Layer            | Components                                                          | Role in the solution              |
|------------------|---------------------------------------------------------------------|-----------------------------------|
| Data layer       | ERP, MES, WMS, QMS, lab, supplier, and channel feeds                | Creates one operational truth     |
| AI layer         | Forecasting, anomaly detection, formula intelligence, vision models | Turns data into decisions         |
| Workflow layer   | Alerts, approvals, exception handling, and task routing             | Makes AI usable by teams          |
| Governance layer | Audit trails, access control, validation, monitoring                | Keeps the system safe and trusted |

## FINAL TAKEAWAY

This case study shows how a beauty manufacturing company can move from fragmented planning and manual quality checks to an AI-assisted operating model that is easier to manage, easier to scale, and easier to trust.

The research points to the same conclusion across demand planning, formulation, sourcing, and packaging: the biggest wins come from clean data, disciplined integration, and human-approved decision-making. That is where AI becomes a practical supply-chain advantage rather than a marketing claim.

## RESEARCH REFERENCES USED FOR THE CASE STUDY

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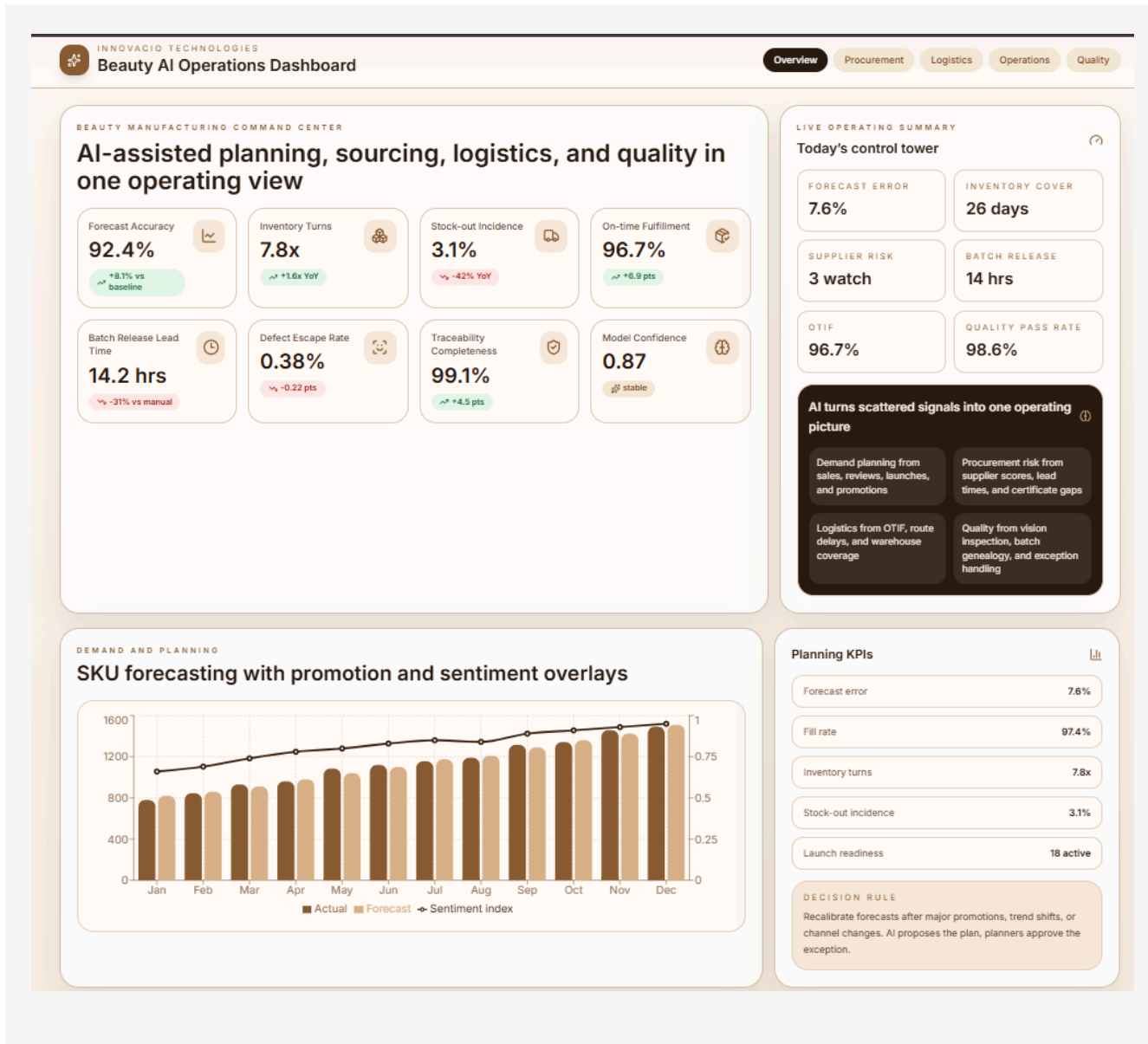
Eppler, A. R., et al. (2025). Artificial Intelligence Beauty Revolution - State of the Art and New Trends from the SCC78 Annual Meeting.

## APPENDIX: KPI SET YOU CAN REUSE

These metrics help you present the problem, the AI intervention, and the result in one clean narrative.

| KPI                       | Why it matters                                | Good evidence source       |
|---------------------------|-----------------------------------------------|----------------------------|
| Forecast error / MAPE     | Shows whether planning improved               | Demand model output        |
| Inventory turns           | Shows whether stock is being held efficiently | ERP / warehouse records    |
| Stock-out incidence       | Connects planning to service                  | Order fulfillment logs     |
| Batch release lead time   | Shows whether quality workflow is faster      | QMS records                |
| Defect escape rate        | Shows packaging and QC improvement            | Quality / returns data     |
| Traceability completeness | Shows source transparency                     | Supplier and batch records |

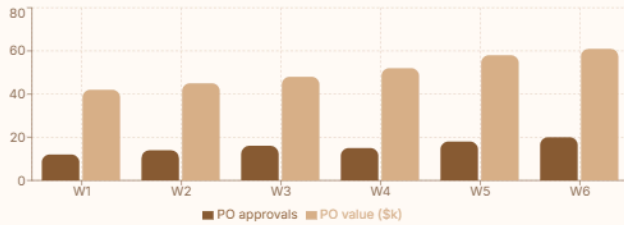
## SCREENSHOTS



PROCUREMENT AND SOURCING

## Supplier intelligence, raw-material risk, and purchase-order flow

### Weekly procurement activity



### Supplier risk ranking

|                                                                     |         |
|---------------------------------------------------------------------|---------|
| <b>GreenRoot Naturals</b><br>Lead time 12 days · Certification 100% | Risk 91 |
| <b>HerbalPure Labs</b><br>Lead time 14 days · Certification 98%     | Risk 87 |
| <b>Botanica Fields</b><br>Lead time 19 days · Certification 92%     | Risk 74 |
| <b>Oceanic Ingredients</b><br>Lead time 16 days · Certification 97% | Risk 81 |
| <b>EarthGlow Extracts</b><br>Lead time 23 days · Certification 88%  | Risk 68 |
| <b>PureSense Actives</b><br>Lead time 11 days · Certification 100%  | Risk 95 |

#### PROCUREMENT LEAD TIME

**12.8 days**

Sensitive ingredients are prioritized first.

#### SPEND AT RISK

**8.4%**

Watchlist suppliers trigger alternate sourcing.

LOGISTICS AND FULFILLMENT

## Warehouse coverage, route performance, and order promise control

#### RAW INVENTORY

**26 days**

Enough for current production cadence

#### WIP INVENTORY

**11 days**

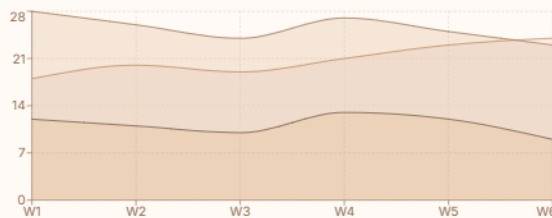
Balanced between mixing and filling

#### FINISHED GOODS

**20 days**

Held for retail and DTC service levels

### Inventory coverage trend



### Fulfillment control

#### OTIF

**96.7%**

#### ROUTE DELAY

**6.3 hrs**

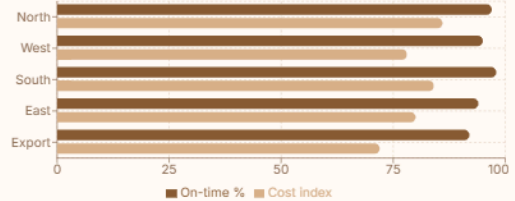
#### PICK ACCURACY

**99.2%**

#### BACKORDERS

**1.8%**

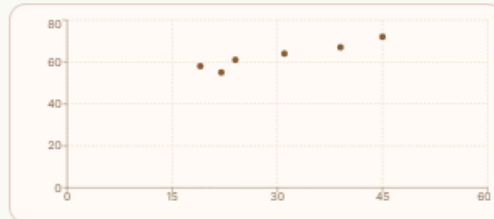
### Regional route health





#### OPERATIONS AND RELIABILITY

### Predictive maintenance for mixers, fillers, cappers, and labelers



#### Line health summary

|                                                |          |
|------------------------------------------------|----------|
| <b>Mixer A</b><br>Vibration 24 - Temp 61       | Risk 9%  |
| <b>Homogenizer B</b><br>Vibration 39 - Temp 67 | Risk 22% |
| <b>Filler C</b><br>Vibration 19 - Temp 58      | Risk 0%  |
| <b>Capper D</b><br>Vibration 31 - Temp 64      | Risk 16% |
| <b>Labeler E</b><br>Vibration 42 - Temp 72     | Risk 27% |
| <b>Cartoner F</b><br>Vibration 22 - Temp 55    | Risk 0%  |

#### UNPLANNED DOWNTIME

**4.2 hrs**

Early warning from sensor drift and work orders

#### MAINTENANCE INTERVENTIONS

**7**

Planned before line failure occurs

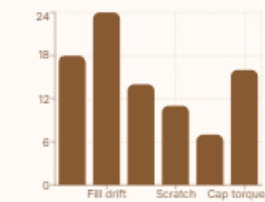
#### OPERATIONAL PRINCIPLE

Start with the most critical assets, reuse PLC and maintenance data, and let the model recommend a maintenance window during planned downtime.

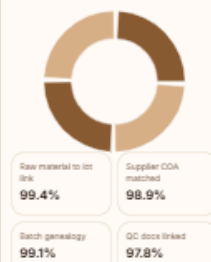
#### QUALITY, COMPLIANCE, AND TRACEABILITY

### Packaging inspection, defect analysis, and batch genealogy

#### Packaging defects by type



#### Traceability completeness

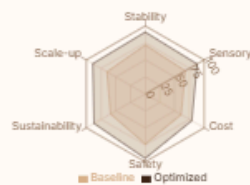


#### Quality gate

|                      |       |                    |       |
|----------------------|-------|--------------------|-------|
| Packaging pass rate  | 98.6% | First-pass release | 91.2% |
| Escapes to customers | 0.38% | Open CAPA          | 2     |

#### Formulation intelligence

#### Optimization radar



|                                                 |            |
|-------------------------------------------------|------------|
| <b>Hydrating serum base</b><br>AI score 93      | Approved   |
| <b>Barrier cream emulsion</b><br>AI score 88    | Plan       |
| <b>SPF stabilizer blend</b><br>AI score 81      | Lab review |
| <b>Botanical cleanser system</b><br>AI score 90 | Approved   |
| <b>Anti-frizz hair mist</b><br>AI score 84      | Plan       |

#### PROCUREMENT LINK

Ingredient selection is linked to supplier quality data, certification, and historical batch performance so formulation decisions stay connected to sourcing reality.

